SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5 **Total Marks:50**

Annexure A

CONCEPT MAPPING

Code of Conduct:

I C ESHWANTH(192321062) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Eshwanth C

Annexure A

CONCEPT MAPPING

1. Create a concept map linking the following: Security Attacks, Threats, Risks and Security Goals (Confidentiality, Integrity, Availability. Perform the following tasks:
   a. Show how each attack leads to specific threats and risks
   b. Map how security goals help mitigate them Provide one real-world example for each link

2. Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Perform the following tasks to connect based on: Key usage, Complexity and Security strength. Analyze which technique is most secure and why.

3. Construct a concept map for Classical Encryption Schemes: Substitution Techniques and Transposition Techniques. Perform the following tasks:
   a. Show differences and similarities
      i. Map how each affects the Data structure and Security level ii. Analyze which is more vulnerable to attacks and why

4. Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks:
   a. Show relationships between these concepts
   b. Analyze how each contributes to secure communication
   c. Include one real-world application for each

5. Draw a concept map linking number theory concepts used in cryptography: Modular Inverse, Extended Euclid Algorithm, Fermat's Little Theorem, Euler's Phi Function and Euler's Theorem. Perform the following tasks:
   a. Show how each concept is mathematically connected
   b. Map their role in encryption/decryption
   c. Analyze which concept is most critical for public key cryptography and justify

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

CONCEPT MAP – APPLICATIONS OF CRYPTOGRAPHY

Cryptography is the practice of securing information by converting it into an unreadable format to ensure confidentiality, integrity and authentication.

GOALS OF CRYPTOGRAPHY

- Confidentiality
- Integrity
- Authentication
- Non-repudiation

1. INTERCEPTED SUSPICIOUS MESSAGE

Cipher Text : GSRH RH Z HVXIVG

a. Likely Original Message

Atbash Cipher (Reverse Alphabet)

THIS IS A SECRET

Atbash Mapping

A ↔ Z
B ↔ Y
C ↔ X
…
M ↔ N

b. Justification

- Atbash maps each letter to its opposite in the alphabet (A↔Z, B↔Y, …).
- Decoding gives meaningful English words.
- Repeating patterns are preserved (RH → IS, HVXIVG → SECRET).

Key Takeaway:
Simple substitution ciphers can be broken using pattern recognition and frequency analysis.

2. SECURE MESSAGING SYSTEM

Technique : Vigenère Cipher (Keyword Based)
Plain Text : HELLO Keyword : KEY

a. Encryption Process (Step-by-Step)

Plain Text	H	E	L	L	O
Keyword	K	E	Y	K	E
P (0-25)	7	4	11	11	14
K (0-25)	10	4	24	10	4
(P+K) mod 26	17	8	9	21	18
Cipher Text	R	I	J	V	S

Keyword Pattern KEY KE

Encrypted Message : RIJVS

b. Role of Repeating Keyword

- Keyword repeats to match the length of the message.
- Each keyword letter determines the shift for the corresponding plaintext letter.
- Different shifts for different letters → higher security than simple substitution.

Key Takeaway:
Keyword-based ciphers increase security by using variable shifts instead of a fixed shift.

3. SUSPICIOUS DIGITAL ASSET INVESTIGATION

A company logo image may contain hidden confidential information.

a. Detecting Hidden Data

- Steganalysis: Analyze pixel patterns & LSB values.
- Metadata Analysis: Check file properties for hidden comments or info.
- Statistical Analysis: Look for abnormal noise, pixel distribution, or compression artifacts.
- Forensic Tools: Use specialized software to detect embedded messages or files.

Image
↓
Hidden Data

b. Techniques to Strengthen Security

1. Encrypt the secret data before embedding it.
   Even if extracted, it remains unreadable without the key.
2. Randomized / Adaptive Embedding.
   Hide data in random pixel locations instead of sequential order to resist detection.

Key Takeaway:
Steganography hides the existence of data.
Strong encryption + smart embedding = better security.

COMMON PRINCIPLES

- Transform data to hide meaning from unauthorized users.
- Use keys, patterns, or structures for encryption.
- Strong security = Strong algorithm + Proper implementation

4. LEGACY ENCRYPTION IN COMMUNICATION

Method : Rail Fence Cipher (3 rails)
Cipher Text : WECRLTEERDSOEEFEAOC[illegible]IVDEN

a. Decryption (Reversing the Process)

Rail 1	W	E	R	T	E	R	O	E	E	E	O	A	D	E
Rail 2	E	C	L	E	R	D	S	E	E	E	A	C	I	N
Rail 3	·	R	E	O	F	E	E	-	A	I	·	·	V	N

Original Message : WE ARE DISCOVERED FLEE AT ONCE

b. Why This Method is Vulnerable?

- Only rearranges letters, does not change them.
- Very small key space (depends on number of rails).
- Patterns can be identified easily.
- Easily broken by trial & error and frequency analysis.

c. Enhancement

Use modern encryption algorithms like AES-256 for strong confidentiality and protection against brute-force attacks.

Key Takeaway:
Transposition ciphers like Rail Fence are not secure in modern systems; use strong encryption algorithms instead.

Overall Learning: Cryptography and Steganography are essential for secure communication and data protection. Choosing the right technique and implementing it properly ensures confidentiality in the digital world.